

Part 1: Understanding MCP

- What problem does MCP solve?
 - MCP solves the problem of inconsistent or incomplete context in AI pipelines.
- How does MCP differ from traditional APIs?
 - MCP server is an interface for LLMs to access tools and data, while traditional APIs are interfaces for humans through applications to access data and functionalities.
- What are common use cases in DevOps?
 - CI/CD automation
 - Efficient code management
 - Infrastructure automation
 - Incident response

Part 2: Setting Up an MCP Server

1. Create a project

```
Last login: Fri Apr 10 03:22:56 on ttys000
jane@Janes-MacBook-Air ~ % mkdir mcp-server-lab
cd mcp-server-lab
[npm init -y
Wrote to /Users/jane/mcp-server-lab/package.json:

{
  "name": "mcp-server-lab",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}
```

2. Install dependencies:

```
[jane@Janes-MacBook-Air mcp-server-lab % npm install express

added 65 packages, and audited 66 packages in 1s

22 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

3. Create `server.js`:

```
[jane@Janes-MacBook-Air mcp-server-lab % cat > server.js
const express = require('express');
const axios = require('axios');

const app = express();
const PORT = 3000;

app.get('/context', async (req, res) => {
  const username = req.query.user || 'github';

  try {
    const response = await axios.get(`https://api.github.com/users/${username}`);

    res.json({
      message: "Hello from MCP Server with external API integration",
      timestamp: new Date(),
      requested_user: username,
      githubData: {
        login: response.data.login,
        name: response.data.name,
        public_repos: response.data.public_repos,
        followers: response.data.followers,
        following: response.data.following,
        profile_url: response.data.html_url
      }
    });
  } catch (error) {
    res.status(500).json({
      error: "Failed to fetch data from external API",
      details: error.message
    });
  }
});

app.listen(PORT, () => {
  console.log(`MCP Server running on port ${PORT}`);
});^D
```

4. Run

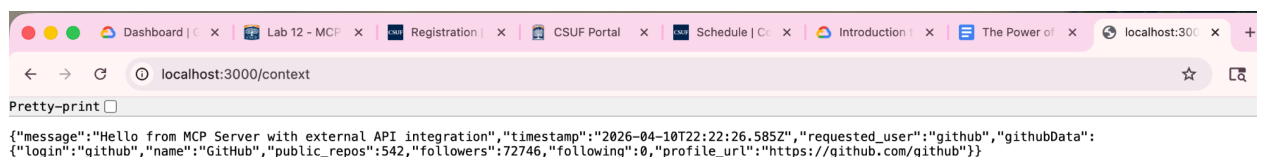
```
[jane@Janes-MacBook-Air mcp-server-lab % npm install axios
up to date, audited 77 packages in 224ms

24 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
[jane@Janes-MacBook-Air mcp-server-lab % node server.js
MCP Server running on port 3000
```

Part 3: Connect MCP Server

- Use a client (CLI or tool) to connect to: <http://localhost:3000/context>

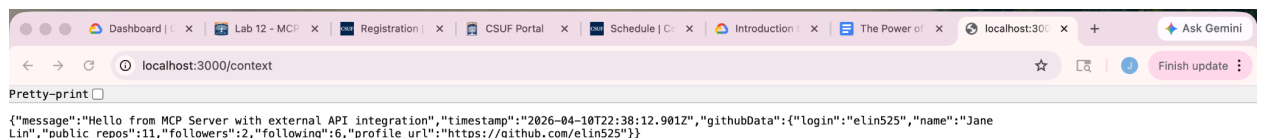


- Test using: `curl http://localhost:3000/context`

```
Last login: Fri Apr 10 15:07:52 on ttys000
[jane@Janes-MacBook-Air mcp-server-lab % curl http://localhost:3000/context
{"message":"Hello from MCP Server with external API integration","timestamp":"2026-04-10T22:24:37.196Z","requested_user":"github","githubData":{"login":"github","name":"GitHub","public_repos":542,"followers":72746,"following":0,"profile_url":"https://github.com/github"}}%
```

Part 4: Extend MCP Server

- Integrate with a public API (weather, GitHub, etc.)



```
jane@Janes-MacBook-Air mcp-server-lab % curl http://localhost:3000/context
{"message":"Hello from MCP Server with external API integration","timestamp":"2026-04-10T22:39:08.482Z","githubData":{"login":"elin525","name":"Jane Lin","public_repos":11,"followers":2,"following":6,"profile_url":"https://github.com/elin525"}}%
```

```
JS server.js X
JS server.js > ...
1  const express = require('express');
2  const axios = require('axios');
3
4  const app = express();
5  const PORT = 3000;
6
7  app.get('/context', async (req, res) => {
8    try {
9      const response = await axios.get('https://api.github.com/users/elin525');
10
11      res.json({
12        message: "Hello from MCP Server with external API integration",
13        timestamp: new Date(),
14        githubData: {
15          login: response.data.login,
16          name: response.data.name,
17          public_repos: response.data.public_repos,
18          followers: response.data.followers,
19          following: response.data.following,
20          profile_url: response.data.html_url
21        }
22      });
23    } catch (error) {
24      res.status(500).json({
25        error: "Failed to fetch data from external API",
26        details: error.message
27      });
28    }
29  });
30
31  app.listen(PORT, () => {
32    console.log(`MCP Server running on port ${PORT}`);
33  });
```

Part 5: DevOps Integration

- Containerize the MCP server using Docker

```
jane@Janes-MacBook-Air mcp-server-lab % cat <<'EOF' > Dockerfile
FROM node:18

WORKDIR /app

COPY package*.json ./
RUN npm install

COPY . .

EXPOSE 3000

CMD ["node", "server.js"]
EOF
jane@Janes-MacBook-Air mcp-server-lab % cat <<'EOF' > .dockerignore
[node_modules
npm-debug.log
.DS_Store
.git
.gitignore
EOF
```

```

[jane@Janes-MacBook-Air mcp-server-lab % docker build -t mcp-server-lab .
[+] Building 12.0s (11/11) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile                0.0s
[=> => transferring dockerfile: 153B                               0.0s
=> [internal] load metadata for docker.io/library/node:18         2.9s
=> [auth] library/node:pull token for registry-1.docker.io        0.0s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 93B                                     0.0s
=> [1/5] FROM docker.io/library/node:18@sha256:c6ae79e38498325db67193d39 7.3s
=> => resolve docker.io/library/node:18@sha256:c6ae79e38498325db67193d39 0.0s
=> => sha256:28b4958639f99536cfbcb306d031319fd840236d205945a 447B / 447B 0.2s
=> => sha256:d757d01e4983550a8a8861a19fe0a4356d8f3be62 45.73MB / 45.73MB 1.5s
=> => sha256:354ffb7f5514f32472b48da285613641ab5a8407968 1.25MB / 1.25MB 0.4s
=> => sha256:4c4eb3d16508b3a9f96749a2d76662a53ec4f171cfa 3.33kB / 3.33kB 0.3s
=> => sha256:b58ee5cb7152015437e4a9b3066ae9e25a26a3b 202.76MB / 202.76MB 4.7s
=> => sha256:1f4f297e4f699ae0f384d5cc1ea42065f58a115aa 64.36MB / 64.36MB 2.2s
=> => sha256:280bbe393e788ced1dcb033580604b24de0836016 23.55MB / 23.55MB 0.6s
=> => sha256:1a12b4ea7c0ce04aa0e98be0a8c9942162bac7142 48.33MB / 48.33MB 1.0s
=> => extracting sha256:1a12b4ea7c0ce04aa0e98be0a8c9942162bac71426f734fe 0.5s
=> => extracting sha256:280bbe393e788ced1dcb033580604b24de083601624337be 0.2s
=> => extracting sha256:1f4f297e4f699ae0f384d5cc1ea42065f58a115aa0a634d4 0.6s
=> => extracting sha256:b58ee5cb7152015437e4a9b3066ae9e25a26a3bef6617d0b 1.8s
=> => extracting sha256:4c4eb3d16508b3a9f96749a2d76662a53ec4f171cfaeb773 0.0s
=> => extracting sha256:d757d01e4983550a8a8861a19fe0a4356d8f3be6255059f6 0.6s
=> => extracting sha256:354ffb7f5514f32472b48da285613641ab5a84079687c376 0.0s
=> => extracting sha256:28b4958639f99536cfbcb306d031319fd840236d205945a6 0.0s
=> [internal] load build context                                  0.0s
=> => transferring context: 35.34kB                                0.0s
=> [2/5] WORKDIR /app                                           0.2s
=> [3/5] COPY package*.json ./                                  0.0s
=> [4/5] RUN npm install                                         1.2s
=> [5/5] COPY . .                                                0.0s
=> exporting to image                                           0.3s
=> => exporting layers                                           0.2s
=> => exporting manifest sha256:c3e944aad01d7df5a460fa3dfc4c602e5a23119b 0.0s
=> => exporting config sha256:db68adb016cc87355a61325896d181abde903df4dc 0.0s
=> => exporting attestation manifest sha256:9a21fd001fe2f5654017a63d4613 0.0s
=> => exporting manifest list sha256:b4c6251884b98c6bede1f59b72004f5a758 0.0s
=> => naming to docker.io/library/mcp-server-lab:latest         0.0s
=> => unpacking to docker.io/library/mcp-server-lab:latest      0.1s

```

```

jane@Janes-MacBook-Air mcp-server-lab % docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	In Use
204312534875.dkr.ecr.us-west-1.amazonaws.com/my-ecr-repo:latest	1c6fd52f0c2b	369MB	106MB	
204312534875.dkr.ecr.us-west-1.amazonaws.com/my-ecrrepo:latest	082377b8dd7d	369MB	106MB	
204312534875.dkr.ecr.us-west-1.amazonaws.com/my-nginx-proxy:latest	c722ea436d32	62.9MB	62.9MB	
mcp-server-lab:latest	b4c6251884b9	1.57GB	389MB	
my-ecr-repo:latest	1c6fd52f0c2b	369MB	106MB	
my-ecrrepo:latest	082377b8dd7d	369MB	106MB	
my-nginx-proxy:latest	c722ea436d32	62.9MB	62.9MB	

```
jane@Janes-MacBook-Air mcp-server-lab % curl "http://localhost:3000/context?user=elin525"
{"message":"Hello from MCP Server with external API integration","timestamp":"2026-04-10T22:52:59.299Z","githubData":{"login":"elin525","name":"Jane Lin","public_repos":11,"followers":2,"following":6,"profile_url":"https://github.com/elin525"}}%
```

- EC2

Dashboard
AWS Global View
Events
Instances
Instances Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing

EC2

Instances

i-008090ff9c9ea47fb

Instance summary for i-008090ff9c9ea47fb (lab12)

Updated 14 minutes ago

Instance ID

i-008090ff9c9ea47fb

IPv6 addresses

–

Hostname type

IP name: ip-172.30.0.194.us-west-1.compute.internal

Answer private resource DNS name

–

Auto-assigned IP address

ip-172.30.0.194.us-west-1.compute.internal

IAM role

–

IMDSv2

Required

Operator

–

Public IPv4 address

54.193.95.150 | open address

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172.30.0.194.us-west-1.compute.internal

Instance type

t3.micro

VPC ID

vpc-0b66c7f8211a600c1

Subnet ID

subnet-04108e3abcd5e94d9

Instance ARN

arn:aws:ec2:us-west-1:204312534875:instance/i-008090ff9c9ea47fb

Private IPv4 addresses

172.30.0.194

Public DNS

ec2-54-193-95-150.us-west-1.compute.amazonaws.com | open address

Elastic IP addresses

–

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

–

Managed

false

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Monitoring details

Monitoring disabled

Allowed image

–

Launch time

Fri Apr 10 2026 16:08:52 GMT-0700 (Pacific Daylight Time) (4 minutes)

Platform details

Linux/UNIX

Termination protection

Disabled

AMI location

amazon/al2023-ami-2023.11.20260406.2-kernel-6.1-x86_64

```
jane@Janes-MacBook-Air mcp-server-lab % ssh -i ~/Downloads/lab12-key.pem ec2-user@54.193.95.150
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@      WARNING: UNPROTECTED PRIVATE KEY FILE!      @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
Permissions 0644 for '/Users/jane/Downloads/lab12-key.pem' are too open.
It is required that your private key files are NOT accessible by others.
This private key will be ignored.
Load key '/Users/jane/Downloads/lab12-key.pem': bad permissions
ec2-user@54.193.95.150: Permission denied (publickey, gssapi-keyex, gssapi-with-mic).
jane@Janes-MacBook-Air mcp-server-lab % chmod 400 ~/Downloads/lab12-key.pem
jane@Janes-MacBook-Air mcp-server-lab % ssh -i ~/Downloads/lab12-key.pem ec2-user@54.193.95.150
```

```
#  
#####  
#  
#####  
##  
V-->  
  
-.-  
_/_/_/_/  
m/'
```

Amazon Linux 2023

<https://aws.amazon.com/linux/amazon-linux-2023>

```
ec2-user@ip-172-30-0-194 ~]$ sudo dnf update -y
sudo dnf install -y docker
sudo systemctl start docker
sudo systemctl enable docker
sudo usermod -aG docker ec2-user
newgrp docker
docker --version
Amazon Linux 2023 Kernel Livepatch repository    266 kB/s | 31 kB    00:00
Last metadata expiration check: 0:00:01 ago on Fri Apr 10 23:20:50 2026.
Dependencies resolved.
Nothing to do.
Complete!
Last metadata expiration check: 0:00:02 ago on Fri Apr 10 23:20:50 2026.
Dependencies resolved.

=====
Package                Arch      Version                               Repository      Size
=====
Installing:
  docker                x86_64    25.0.14-1.amzn2023.0.2              amazonlinux     46 M
```

```
Last login: Fri Apr 10 23:19:06 2026 from 47.145.158.167
[ec2-user@ip-172-30-0-194 ~]$ cd ~/mcp-server-lab
docker build -t mcp-server-lab .
docker run -d -p 3000:3000 --name mcp-server mcp-server-lab
docker ps
[+] Building 20.8s (7/9)
=> [internal] load build definition from Dockerfile                                docker:default    0.0s
=> => transferring dockerfile: 212B                                              0.0s
=> [internal] load metadata for docker.io/library/node:18                        1.0s
-----
410b63b00a118a2168257148c875aca547e89e9001351c6febfc3e5f01dc402
CONTAINER ID   IMAGE                                COMMAND                  CREATED          STATUS          PORTS                               NAMES
410b63b00a11   mcp-server-lab                      "docker-entrypoint.s-"   1 second ago    Up Less than a second    0.0.0.0:3000->3000/tcp, :::3000->3000/tcp    mcp-server
[ec2-user@ip-172-30-0-194 mcp-server-lab]$ curl "http://54.193.95.150:3000/context?user=elin525"
{"message":"Hello from MCP Server with external API integration","timestamp":"2026-04-10T23:32:30.012Z","githubData":{"login":"elin525","name":"Jane Lin","public_repos":11,"followers":2,"following":6,"profil
```

```
[ec2-user@ip-172-30-0-194 mcp-server-lab]$ █
```